
1. Sales of a Retail Store

Observation Table

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Sales (Y)	150	152	155	160	166	173	181	190	200	212

Question

Fit a trend using:

- a) free-hand curve
- b) semi-average
- c) 3-year moving average
- d) 4-year central moving average
- e) 5-year moving average
- f) 7-year moving average

2. Steel Production

Observation Table

Year	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
Production (Y)	450	455	460	470	485	500	520	540	565	590

Question

Apply methods (a–f) as above.

3. National Park Visitors

Observation Table

Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
Visitors (Y)	300	305	310	315	322	330	340	352	365	380

Question

Calculate trends using methods (a–f).

4. Electricity Demand

Observation Table

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	
Demand (Y)	900	910	925	940	960	985	1010	1040	1075	1115	

Question

Fit trend using (a–f).

5. Registered Vehicles

Observation Table

Year	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	
Vehicles (Y)	520	530	545	560	580	600	625	650	680	715	

Question

Estimate trend via methods (a–f).

6. Milk Production

Observation Table

Year	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	
Production (Y)	120	123	127	132	138	145	153	162	172	183	

Question

Apply trend fitting methods (a–f).

7. Export Values

Observation Table

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	
Export (Y)	800	815	830	850	875	900	930	965	1005	1050	

Question

Compute trend using (a–f).

8. Company Employees

Observation Table

| Year | 2001 | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 |
| Employees (Y) | 45 | 47 | 49 | 52 | 55 | 59 | 63 | 68 | 74 | 81 |

Question

Fit trends using methods (a–f).

9. Rice Production

Observation Table

| Year | 1995 | 1996 | 1997 | 1998 | 1999 | 2000 | 2001 | 2002 | 2003 | 2004 |
| Production (Y) | 250 | 255 | 262 | 270 | 280 | 292 | 305 | 320 | 336 | 353 |

Question

Find trend using methods (a–f).

10. Town Population

Observation Table

| Year | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
| Population (Y) | 95 | 97 | 100 | 104 | 109 | 115 | 122 | 130 | 139 | 149 |

Question

Fit trend using:

- a) free-hand
- b) semi-average
- c) 3-year moving average
- d) 4-year central moving average
- e) 5-year moving average
- f) 7-year moving average